ISTA 130 Execution Trace and Functions Worksheet

Write a function called `diff_squares` that takes two numbers as arguments and prints the square of the first number minus the square of the second one. An example call:

```
>>> diff_squares(7, 13)
The difference between the squares of 7 and 13 is: -120
```

Consider the following function:

```
def sum_squares(max):
    ''' Print the sum of squares of integers 1 to max. '''
    sum = 0
    print('x - x**2')
    print('--------')
    for i in range(max):
        num = i + 1
        square = num**2
        print(num, '-', square)
        sum = sum + square
    print()
    print('Sum of squares for 1 through', max, ":", sum)
```

What is the output of the function call `sum_squares(3)`? I suggest making a table of the variables and keeping track of their values as the code executes (this is an execution trace).

(Continued on other side)

Write a function called `letter_square` that takes two arguments: a character and a size a. It prints a line with the character repeated size times. It then prints size – 2 lines with the character followed by size – 2 spaces followed by the character. It then repeats the first line. An example call:

```
>>> letter_square('W', 7)
WWWWWWW
W     W
W     W
W     W
W     W
W     W
WWWWWWW
```

You can use string multiplication and addition. An example:

```
>>> '$' * 3 + '-' * 5 + '$' * 3
'$$$-----$$$'
```